

1. Administrative & Study Identification

Full Study Title: Study to Assess Safety, Efficacy, and Cellular Kinetics of YTB323 in Generalized Myasthenia Gravis **Short Title/ Acronym:** YTB323 in gMG **Protocol Number:** CYTB323012101 **NCT Number:** NCT06704269 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** YTB323 (Autologous CD19-directed CAR-T cell therapy) **Phase of Development:** Phase I/II **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, efficacy, and cellular kinetics of YTB323 in participants with treatment-resistant generalized myasthenia gravis, specifically observing the occurrence, severity, and frequency of adverse events (AEs) and serious AEs. **Secondary Objectives:** To evaluate preliminary efficacy outcomes, including change in the Myasthenia Gravis-Activities of Daily Living (MG-ADL) score, Quantitative Myasthenia Gravis (QMG) total score, and the proportion of patients achieving a Post-Intervention Status of minimal manifestations or better.

2.2 Background & Rationale To address the high burden of disease and clinical gap in patients with treatment-resistant generalized myasthenia gravis (gMG). B cells play a major role in gMG by producing autoantibodies like anti-AChR and anti-MuSK. YTB323 is an autologous CD19-directed chimeric antigen receptor (CAR) T-cell therapy designed to specifically deplete these autoreactive B cells. This one-time investigational treatment aims to induce prolonged disease control and significantly reduce or eliminate the need for chronic non-steroidal immunosuppressants.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Open-label, consisting of a sentinel cohort followed by an expansion cohort) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target \$N\$:** 15 participants (3 in the sentinel cohort, 12 in the expansion cohort) **Number of Sites:** Multicenter (Includes sites in the United States and France) **Estimated Study Start:** 22-Apr-2025 **Estimated Primary Completion:** 26-Oct-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 18 to 65 years. 2. Confirmed gMG diagnosis supported by documented positive serology testing for either AChR antibodies or MuSK antibodies. 3. Treatment-resistant gMG as defined by an MG-ADL score ≥ 6 ($\geq 50\%$ non-ocular) at screening despite adequate treatment with at least two different non-steroidal immunosuppressive drugs.

4.2 Exclusion Criteria 1. Prior treatment with anti-CD19 therapy, adoptive T cell therapy, or any prior gene therapy product (e.g., CAR-T cell therapy). 2. Other uncontrolled disease states (e.g., asthma, inflammatory bowel disease) where flares are commonly treated with systemic corticosteroids. 3. Known immunodeficiency syndrome or tested positive for HIV antibody.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single-dose administration **Route of Administration:** Intravenous infusion **Duration of Treatment:** Single administration of YTB323, followed by a mandatory 15-year Long-Term Follow-up (LTFU) period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of cholinesterase inhibitors (for ≥ 2 weeks). Corticosteroids are permitted if the patient is able to taper to a maximum dose of 10 mg prednisolone daily (or equivalent) at least one week before leukapheresis. **Prohibited:** Concurrent CD19 targeted therapies, biological immunomodulators, or gene therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures	:---	:---
:---	Screening	Days -30 to 0	Informed Consent, Serology testing, MG-ADL scoring, Corticosteroid tapering	Baseline
Day 0	Leukapheresis, Autologous CAR-T manufacturing, YTB323 Infusion	Interim	Post-Infusion	Intensive inpatient/
				outpatient safety monitoring for CRS and ICANS, Efficacy scaling
End of Study	Year 15	Final Long-Term Follow-Up (LTFU)		assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Occurrence, severity, and frequency of adverse events (AEs) and serious AEs, notably Cytokine Release Syndrome (CRS) and Immune Effector Cell-Associated Neurotoxicity Syndrome (ICANS). **Measurement Method:** Clinical observation, standard AE toxicity scales, and vital sign monitoring.

7.2 Secondary & Exploratory Endpoints * Change from baseline in the Myasthenia Gravis-Activities of Daily Living (MG-ADL) total score and Quantitative Myasthenia Gravis (QMG) total score. * Proportion of patients achieving a Myasthenia Gravis Foundation of America Post-Intervention Status (MGFA-PIS) of minimal manifestations or better, sustained for 6 months. * Assessment of plasma pharmacokinetics and cellular immunogenicity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring for CRS, neurological toxicities (ICANS), and changes in systemic inflammatory parameters. **Serious Adverse Events (SAEs):** Evaluated strictly throughout the primary observation period and extending into the 15-year LTFU. **Data Monitoring Committee (DMC):** Independent safety reviews are utilized to assess sentinel cohort (first 3 patients) safety data before permitting dosing in the expansion cohort (12 patients).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all participants exposed to YTB323; Efficacy Set for valid MG-ADL and QMG observations. **Sample Size Justification:** N=15, designed as an exploratory, non-confirmatory sample size intended to establish a benefit-to-risk assessment for the future development of YTB323 in gMG. **Handling of Missing Data:** Standard descriptive statistics and imputation techniques designed for Phase I/II exploratory clinical trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring post-infusion, followed by integration into long-term gene therapy tracking registries. **Audit Trail:** Rigorous tracking of CAR-T cell "chain of identity" from leukapheresis to infusion, alongside eCRF locking for clinical data.

11. Study Identifiers & Governance

Primary ID: {{CYTB323012101}} **Registry Number:** {{NCT06704269}}
Sponsor/Lead Organization: {{Novartis Pharmaceuticals}} **Study Title:** {{Study to Assess Safety, Efficacy, and Cellular Kinetics of YTB323 in Generalized Myasthenia Gravis}} **Official Regulatory Title:** {{An open-label, multi-center, non-confirmatory study intended to assess safety, efficacy, and cellular kinetics of

YTB323 treatment in participants with treatment-resistant generalized myasthenia gravis}}

12. Clinical Framework (Disease Specific)

Primary Condition: {{Generalized Myasthenia Gravis (gMG)}}
Diagnosis Threshold: {{Positive AChR or MuSK serology and MG-ADL score ≥ 6 }}
Medical Methodology: {{Autologous CD19-directed CAR-T cell therapy (YTB323)}}
Optimization Target: {{Sustained disease control, B-cell depletion, and reduced non-steroidal immunosuppressant use}}

13. Study Procedures & Methodology

Management Pathway: {{Leukapheresis followed by single-dose YTB323 infusion}}
Education Protocol (HCP): {{Specific training on the identification and management of CRS and ICANS}}
Education Protocol (Patient): {{Informed consent regarding the requirement for a 15-year LTFU for gene therapies}}
Quality Audit Mechanism: {{Longitudinal cellular kinetics monitoring & centralized AE tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{15 years (Long-Term Follow-up)}}
Onsite Visit Cadence: {{Intense post-infusion safety monitoring followed by protocol-defined long-term evaluations}}
Remote Monitoring Frequency: {{Regular outcome collection (e.g., MG-ADL via trained evaluator)}}
Checklist Review Interval: {{Ongoing active SAE monitoring post-infusion}}

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			_____			[DD-MMM-YYYY]							
	Clinical Operations Lead			[Insert Name]			_____			[DD-MMM-YYYY]							
	Lead Statistician			[Insert Name]			_____			[DD-MMM-YYYY]							